

## CONTACT


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## EDUCATION

**Information Systems Engineering**
UTN - Tucumán Regional Faculty | 5th year | Apr 2021 - Present

**Full-Stack Development (MERN)**
Rolling Code School | Aug 2023 - May 2024

## TECHNICAL SKILLS

**Backend:**
Node.js, NestJS, Express, Python, FastAPI, TypeScript, Java, Spring Boot

**Frontend:**
React, Next.js, Vite, HTML5, CSS3, Thymeleaf, HTMX

**Data & Databases:**
PostgreSQL, MySQL, MongoDB, Redis, ChromaDB (Vector DB), Prisma, Hibernate/JPA, Flyway

**Cloud & DevOps:**
Proxmox VE, LXC, Docker, OCI, AWS, CI/CD, GitHub Actions, TrueNAS

**Networking & Systems:**
TCP/IP Networking, SDN, Linux System Administration, Nginx, systemd, Tailscale, SSH/TLS, Cisco Packet Tracer

**AI & Automation:**
LangGraph, LangChain, Multi-Agent Systems, RAG, LLMs, Spec-Driven Development (SDD), n8n, BullMQ

**Management & QA:**
Project Management (PMBOK), Scrum, Jira, Automated Testing, Cypress, Postman, Cucumber (BDD), JUnit 5, Testcontainers

## CERTIFICATIONS

**NestJS Backend Developer**
Xetro AI (formerly Vortex)


TUCUMÁN, ARGENTINA


# Mauro Nahuel Armas

### Full-Stack Development | Infrastructure | Automations & AI

Focus on software development and architecture, infrastructure, and business process automation.

## PROFESSIONAL SUMMARY


Fifth-year Information Systems Engineering student and Full-Stack & Cloud Engineer specializing in distributed backend architectures, private cloud infrastructure, and multi-agent generative AI systems. Proven track record of delivering production-grade platforms backed by 900+ automated tests, RAG pipelines, and containerized microservices. Applies PMBOK, Scrum, and Spec-Driven Development (SDD) practices to align complex technical architectures with business process automation.


## EXPERIENCE & ENGINEERING PROJECTS

### 01 NAP - Academic Private Cloud Platform | Cloud Infrastructure & Backend Engineer

UTN-FRT (Supervised Practicum) | Sep 2026 - Nov 2026 | Team of 3 | Paper Accepted at CONAIISI 2026

- Architected an on-demand private cloud platform turning a Proxmox VE and TrueNAS cluster into automated IaaS, PaaS, and SaaS services for university courses.
- Engineered the full lifecycle of request approval, resource reservation, and isolated container provisioning (LXC/Docker/OCI) using Python, FastAPI, and React, secured via SDN/TLS and validated by 900+ automated tests.

**Technologies:** Python, FastAPI, React, Vite, PostgreSQL, Proxmox VE, LXC, Docker, OCI, TrueNAS, SDN, HTTPS, SSH, TLS

### 02 TrimIA - Multi-Agent AI Platform for Business Processes | AI & Full-Stack Engineer

Credimisión S.R.L. (Degree Thesis, UTN-FRT) | Apr 2026 - Present | Individual Development

- Designed and orchestrated a generative AI and RAG multi-agent ecosystem using LangGraph, NestJS, and ChromaDB to centralize WhatsApp customer support, financed sales advising, and internal training.
- Built a real-time React supervision dashboard featuring human-in-the-loop escalation, automated commercial validation, and asynchronous task queues with BullMQ, Redis, and n8n.

**Technologies:** NestJS, TypeScript, React, LangGraph, LLMs, ChromaDB, RAG, PostgreSQL, Prisma, BullMQ, Redis, n8n, WhatsApp API, Docker, Tailscale

### 03 Guardia Médica - ER Clinical Triage System | Backend & QA Automation Developer

UTN-FRT (Software Engineering) | Oct 2025 - Dec 2025 | Team of 3

- Built a hospital admission and triage platform that dynamically prioritizes patients across 5 clinical severity levels and wait times with deterministic FIFO tie-breaking.
- Implemented a 4-layer Clean Architecture in NestJS with strict vital-sign domain validation, JWT/Argon2 authentication, and automated Cucumber BDD test suites in CI.

**Technologies:** NestJS, TypeScript, React, Vite, MySQL 8, Docker, Cucumber BDD, JWT, Argon2

Spec Driven Development (SDD) and AI agents
Udemy

## LANGUAGES

Spanish: native | English: intermediate

### 04 Portal de Oficios Concepción | Full-Stack Developer

Municipality of Concepción & Rolling Code School | Nov 2023 - Mar 2024 | Team of 4

- Developed a municipal web marketplace connecting local residents with verified skilled tradespeople through an administrative moderation and onboarding workflow.
- Implemented RESTful APIs in Express and MongoDB paired with a React frontend, featuring stateless JWT authentication, Cloudinary document storage, and a community review system.

**Technologies:** React, Vite, Express, Node.js, MongoDB, Mongoose, Cloudinary, JWT

### 05 Multi-Tenant Gym Management SaaS Platform | Software Engineer (Java / Spring Boot)

Personal Project (Commercial Client Spec) | Nov 2026 - Present | Individual Development

- Transformed a legacy desktop application into a multi-tenant web platform supporting isolated client data, role-based access control (RBAC), configurable business rules, and custom branding per gym.
- Engineered the backend with Java 21, Spring Boot 3, and PostgreSQL (Flyway) to manage memberships, partial payments, and capacity-limited class bookings, backed by JUnit 5 and Testcontainers in GitHub Actions CI.

**Technologies:** Java 21, Spring Boot 3, Spring Security, JPA, Hibernate, PostgreSQL, Flyway, Thymeleaf, HTMX, JUnit 5, Testcontainers, Docker, GitHub Actions